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The quantum treatment of a diatomic molecule

Consider two particles of masses m_1 and m_2 held together by an attractive central potential energy $V(r)$. The Hamiltonian which describes the motion of about the centre of mass is

$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 + V(r) \tag{C.1}$$

where μ is the reduced mass given by

$$\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}. \tag{C.2}$$

In spherical coordinates the Laplacian operator is

$$\nabla^2 = \frac{\partial^2}{\partial r^2} + 2\frac{\partial}{\partial r} + \frac{1}{r^2}\left(\frac{\partial^2}{\partial \theta^2} + \cot\theta\frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right). \tag{C.3}$$

We can represent the square of the orbital angular momentum operator as

$$\hat{l}^2 = -\left(\frac{\partial^2}{\partial \theta^2} + \cot\theta\frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right). \tag{C.4}$$

The eigenstates of $\hat{l}^2$, which are called spherical harmonic functions $Y_{l,m}(\theta,\phi)$, satisfy the eigenvalue equations

$$\hat{l}^2 Y_{l,m}(\theta,\phi) = l(l+1)Y_{l,m}(\theta,\phi) \tag{C.5}$$

and

$$\hat{l}_z Y_{l,m}(\theta,\phi) = mY_{l,m}(\theta,\phi). \tag{C.6}$$

The single-particle wavefunction can be written as

$$\Phi(r,\theta,\phi) = \frac{R(r)Y_{l,m}(\theta,\phi)}{r} \tag{C.7}$$

where the radial wavefunction $R(r)$ satisfies the differential equation

$$-\frac{\hbar^2}{2\mu}\frac{\mathrm{d}^2R(r)}{\mathrm{d}r^2} + \frac{\hbar^2 l(l+1)R(r)}{2\mu r^2} + V(r)R(r) = \varepsilon R(r). \tag{C.8}$$

This equation is similar to a one-dimensional Schrödinger equation in which the effective potential is

$$U(r) = V(r) + \frac{\hbar^2 l(l+1)}{2\mu r^2}. \tag{C.9}$$

The second term on the right-hand side of eqn (C.9) can be thought of as a centrifugal potential energy.

For a molecule, such as carbon monoxide, the potential $V(r)$ has a deep minimum at a separation $r = d$. To a good approximation we can treat the centrifugal potential energy as if the particles were a constant distance d apart. Then

$$U(r) \approx V(r) + \frac{\hbar^2 l(l+1)}{2\mu d^2}. \tag{C.10}$$

The second term on the right-hand side is the kinetic energy of rotation about the centre of mass. We can treat μd^2 as the moment of inertia, I, of the molecule.

We rewrite eqn (C.8) as

$$-\frac{\hbar^2}{2\mu}\frac{\mathrm{d}^2R(r)}{\mathrm{d}r^2} + \frac{\hbar^2 l(l+1)R(r)}{2\mu d^2} + V(r)R(r) \approx \varepsilon R(r). \tag{C.11}$$

The term $\hbar^2 l(l+1)/2\mu d^2$ just shifts the energy by a constant amount and so it can be removed. The total energy eigenvalue is written as

$$\varepsilon = \varepsilon_n + \frac{\hbar^2 l(l+1)}{2\mu d^2}. \tag{C.12}$$

The solution of the resulting equation then yields a set of eigenstates $R_{n,l}(r)$ and eigenvalues ε_n which correspond to vibrational motion in the potential well. The total energy is the sum of rotational and vibrational energies. For r close to d, the potential varies as $(r-d)^2$, so the vibrational state is that of a simple harmonic oscillator. The energy eigenvalue, $\varepsilon_n = \hbar\omega(n+1/2)$, only depends on n in this approximation. The total energy eigenvalue is

$$\varepsilon_{n,l} = \hbar\omega(n+1/2) + \frac{\hbar^2 l(l+1)}{2\mu d^2}. \tag{C.13}$$

The splitting of the energy levels of rotation is equal to $\hbar^2(l+1)/\mu d^2$ between the l and $(l+1)$ states. The splitting can be observed by spectroscopic techniques. For carbon monoxide this splitting in energy is consistent with the observed spectra if $\hbar/2\pi\mu d^2 = 115.2\,\mathrm{GHz}$.

There are additional complications when the particles in the molecule are identical, as happens in molecular hydrogen. Suppose the two identical atoms are in a state of definite orbital angular momentum, such as

$$\Phi_{n,l,m}(r,\theta,\phi) = \frac{R_{n,l}(r)Y_{l,m}(\theta,\phi)}{r} \tag{C.14}$$

where r is the distance of one atom from the centre of mass. If now $\mathbf{r}$ is changed to $-\mathbf{r}$ the two particles are exchanged. How does the wavefunction change under this transformation?

In spherical coordinates the transformation is equivalent to

$$(r,\theta,\phi) \to (r,\pi-\theta,\pi+\phi). \tag{C.15}$$

Under this transformation the radial function is unchanged whereas the spherical harmonic function transforms as

$$Y_{l,m}(\pi-\theta,\pi+\phi) \to \ (-1)^l\, Y_{l,m}(\theta,\phi). \tag{C.16}$$

In other words the wavefunction changes sign on interchange of particles if l is an odd integer but keeps the same sign if l is an even integer. For example,

$$\begin{aligned} Y_{1,1}(\theta,\phi) &= A\sin(\theta)\,\mathrm{e}^{\mathrm{i}\phi} \to A\sin(\pi-\theta)\,\mathrm{e}^{\mathrm{i}\pi}\,\mathrm{e}^{\mathrm{i}\phi} \\ &= -A\sin(\theta)\,\mathrm{e}^{\mathrm{i}\pi}\,\mathrm{e}^{\mathrm{i}\phi} = -Y_{1,1}(\theta,\phi). \end{aligned}$$

The wavefunction changes sign for $l=1, m=1$.